

Javon Darby

ITAI 2376

Professor Patricia Mcmanus

2/12/2026

## **Arrival (2016): Language and Natural Language Processing**

### **Introduction**

The film *Arrival* (Villeneuve, 2016) explores what happens when humans attempt to communicate with extraterrestrials for the first time. Although it is science fiction, the communication challenges shown in the movie closely relate to real problems in Natural Language Processing (NLP). NLP focuses on how computers understand and process human language.

Language is complex words can have multiple meanings, sentence structures vary, and cultural background affects interpretation. In *Arrival*, Dr. Louise Banks works to understand the alien language, and her experience reflects many of the same challenges researchers face in NLP today.

### **Language and Meaning in the Movie**

When alien spacecraft appear around Earth, Louise Banks, a linguist, is asked to help translate the aliens' language. The aliens, called heptapods, use circular symbols that represent complete ideas rather than linear sentences.

A key moment in the movie occurs when the aliens use a word that humans translate as “weapon.” This causes fear and political tension. Later, Louise realizes the word could also mean “tool” or “gift.” The misunderstanding happens because the meaning depends on context.

This situation reflects a major challenge in NLP: ambiguity.

## **NLP Challenges in *Arrival***

### **Ambiguity**

Ambiguity occurs when a word or sentence has more than one meaning. Jurafsky and Martin (2023) explain that ambiguity is one of the central problems in NLP. In the movie, the word “weapon” is misunderstood because humans interpret it based on their assumptions.

Modern NLP models, such as BERT, attempt to use context to determine meaning (Devlin et al., 2019). However, systems still struggle with sarcasm, tone, and cultural nuance. The movie shows how serious misunderstandings can result from ambiguous language.

### **Structural Differences**

The heptapods’ language is non-linear and circular. It does not follow the left-to-right structure of English. Each symbol expresses a full idea at once.

In real-world NLP, languages differ in grammar and word order. Machine translation systems must account for these differences (Manning & Schütze, 1999). The alien language in the film exaggerates this issue, forcing Louise to rethink her assumptions about how language works.

### **Cultural and Conceptual Barriers**

The military in the film assumes the aliens are hostile. That assumption influences how the message is interpreted. This reflects how bias and perspective affect language understanding.

In NLP, systems are trained on human-created data. If the data contains bias, the model may reflect it (Jurafsky & Martin, 2023). Cultural differences, slang, and regional language can also lead to misinterpretation.

The movie shows that translation is not just about words; it is about understanding perspective.

### **Limited Data**

At the beginning of the film, Louise has very little data. She builds understanding step by step by observing patterns and testing interpretations.

Similarly, NLP systems require large datasets to perform well (Devlin et al., 2019). When data is limited, performance decreases. The process shown in the film—collecting information, forming hypotheses, and refining conclusions—mirrors how machine learning models are trained.

## **Communication Methods and NLP Approaches**

### **Rule-Based NLP**

Louise initially uses structured methods by labeling objects and building vocabulary. This approach resembles early rule-based NLP systems, which relied on grammar rules and dictionaries (Manning & Schütze, 1999). These systems worked in simple cases but struggled with complexity.

### **Statistical NLP**

As more alien symbols are gathered, patterns begin to emerge. This reflects statistical NLP, which relies on probability and word patterns rather than fixed rules (Manning & Schütze, 1999).

### **Deep Learning**

The major breakthrough happens when Louise begins thinking in the alien language. She internalizes its structure. This mirrors modern deep learning models that learn representations of language through neural networks (Devlin et al., 2019).

Deep NLP systems do not rely only on rules; they learn patterns from large amounts of data. Louise's cognitive shift reflects this deeper level of language understanding.

## Reflection

One of the main ideas in *Arrival* is that language shapes perception. This connects to the idea that language influences how people think.

In NLP, this raises an important question: do machines truly understand language, or do they simply predict patterns? Current systems are powerful, but they rely on statistical relationships rather than true comprehension.

The film also highlights responsibility. A single mistranslation nearly leads to global conflict. In real-world applications, NLP errors could affect healthcare, law, or public communication.

Accuracy and context matter.

## Conclusion

*Arrival* (Villeneuve, 2016) presents many of the same challenges faced in Natural Language Processing. These include ambiguity, structural differences, cultural bias, and limited data.

The communication process in the film reflects the evolution of NLP from rule-based systems to statistical models and deep learning. Overall, the movie demonstrates that language is complex and deeply connected to context and perspective.

## References

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, 4171–4186.

Jurafsky, D., & Martin, J. H. (2023). *Speech and language processing* (3rd ed., draft).

<https://web.stanford.edu/~jurafsky/slp3/>

Manning, C. D., & Schütze, H. (1999). *Foundations of statistical natural language processing*. MIT Press.

Villeneuve, D. (Director). (2016). *Arrival* [Film]. Paramount Pictures.